School of Engineering

Course: Diploma in Electronic and Computer Engineering
Diploma in Aerospace Systems & Management

Learning Unit : EG2E55/EG2C68 Microcontroller Programming & Interfacing

Experiment No: Lab 07

Experiment Title: Writing External Interrupt Service Routines

Objective: To learn how to write external interrupt service routine

1. The External Interrupt Circuit

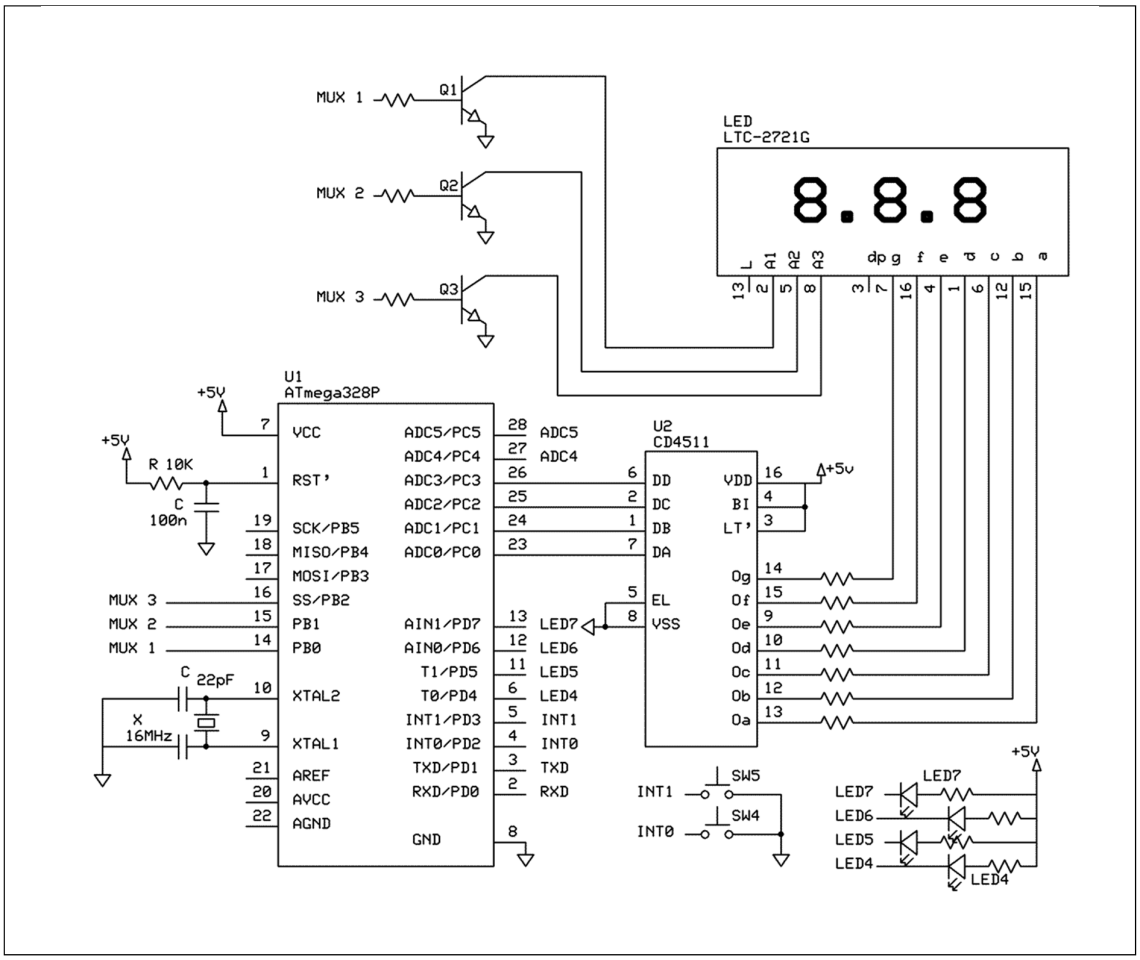


Figure 1-1

2. Polling Method in Response to Switch Input

The program in listing 2-1 makes the last digit of the 3-digit display runs from 0 to 9 repeatedly forever. When the switch connected to PD2 (SW4) is pressed, the set of 4 LEDs blink 5 times (*on + off = 1 time*). The program operation cab be represented in the state diagram shown in Figure 2-1 Compile and execute the program. Observe the operation. You will notice that sometime the program does not respond immediately to the pressing of the switch. Why?

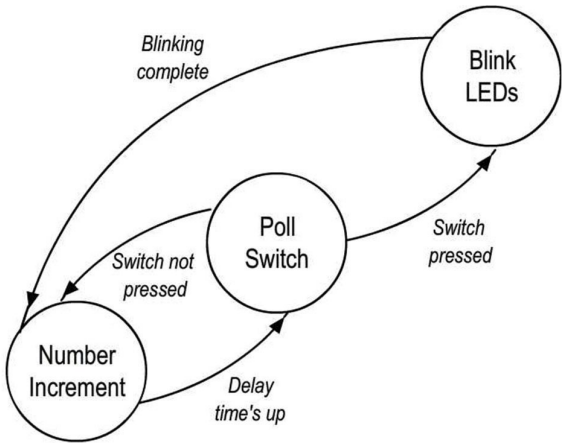


Figure 2-1

. . . . . .

. . . . . .

```
#include <avr/io.h>

unsigned char sec=0;

void delay(unsigned long w){
  while(w--!=0);
}

void blink(){
  unsigned char k;
  for(k=0;k<10;k++){
    PORTD ^= 0xF0; // toggle LEDs.
    delay(100000);
  }
}
```

```
int main(void)
{
  unsigned char x=0;
  DDRD = 0xF0;
  DDRC = 0x0F;
  DDRB = 0x07;
  PORTB |= 0x04; // MUX3 on.
   PORTD &= ~(0xF0);  // Off LEDs.
   PORTD |= 0x0C;    // Set inputs.

  while(1){
    PORTC= (PORTC &0xF0) | x;
    delay(400000);
    x++;
    if (x>9) x=0;  // Reset.

    // Check for switch.
     if ((PIND & 0x04)==0) blink();
  }
}
```

Listing 2-1

3.1 Exercise 1

The program in listing 2-1 can be written using interrupt (see Figure 3-1). The pressing of the switch connected to PD2 (which is also an external interrupt input pin) generates an external interrupt. The interrupt service routine will then handle the blinking of the LED. Complete the interrupt-based program in listing 3-1.

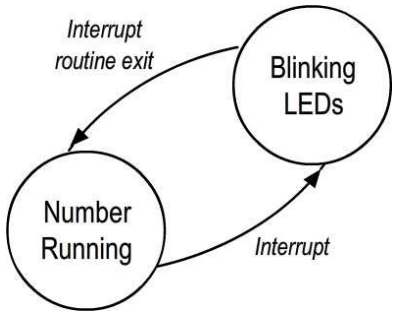


Figure 3-1

```
#include <avr/io.h>
#include <avr/interrupt.h>


void delay(unsigned long w){
  while(w--!=0);
}

// Write the interrupt service
// routine for external interrupt.
```

```
int main(void)
{
  unsigned char x=0;
  DDRD = 0xF0;
  DDRC = 0x0F;
  DDRB = 0x07;
  PORTB |= 0x04; // MUX3 on.
   PORTD &= ~(0xF0);  // Off LEDs.
  PORTD |= 0x0C;    // Set inputs.
  sei();  // Global Interrupt.
  // Set falling edge trigger.
  // Enable interrupt.

  while(1){
    PORTC= (PORTC & 0xF0) | x;
    delay(100000);
    x++;
    if (x>9) x=0;
  }
}
```

Listing 3-1

3.2 Exercise 2

Improve your program in exercise 1 to include a response to interrupt 1 as well. External interrupt 0 shall blink the LED4 & LED5 while external interrupt 1 shall blink the LED6 & LED7. The left most digit of the 3-digit display shall run instead of the right most digit.

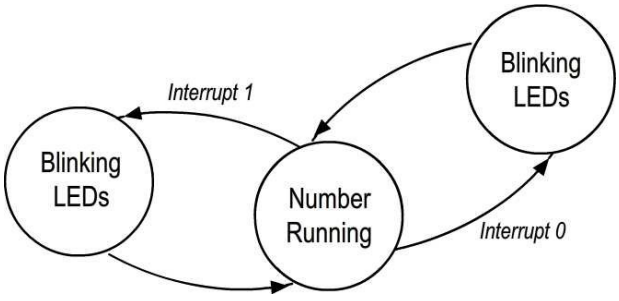
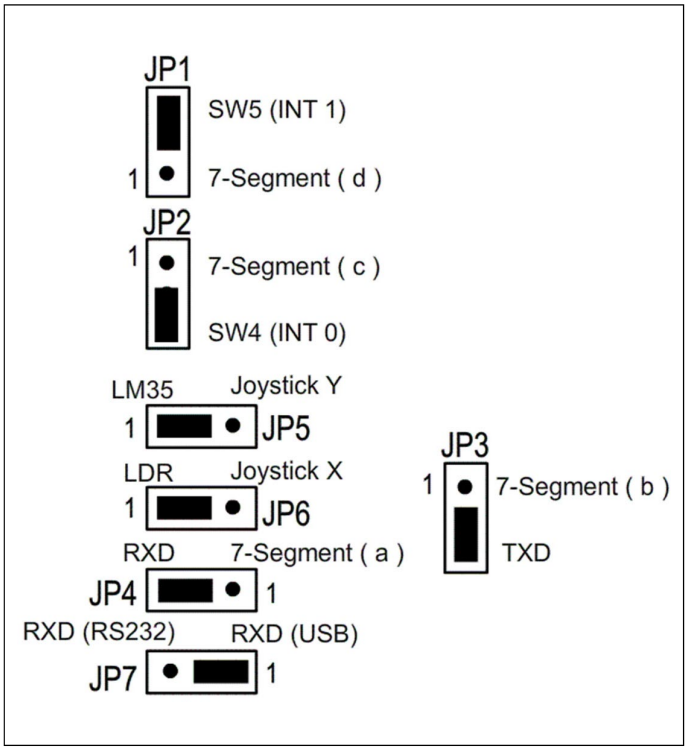


Figure 3-2

Appendix - Circuit Board Configuration



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